

ActInf Textbook Group ~ Cohort 2 ~ Meeting 15 (Chapter 6, part 2)

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Hi, it's February 8th, 2023. We're in meeting 15 of the ACT-INF textbook group. We're in our second discussion in cohort two on chapter six. So last week, we looked at a few questions and in this modeling chapter, there's many questions and topics we can come to. So does anybody want to just raise a question or thought on six? Otherwise, we will look through questions and see what we can do. Yeah, well, I've put a few questions there that came to my mind.

Great. Which one would you like to start with?

So we already talked about this last time, this figure 6.1 on page 110, I believe it is.

And where is it? Page 108. And my question was this vertical connection there between active states and sensory states.

And I was not able to really understand why we would need to model this. And I've put this into the question there.

And that chapter six, should I read out that question? So how is the mutual interaction between active states and sensory states meant?

Can they mutually change their states without impacting neither internal nor external states?

So that means they cycle around. If yes, what is it supposed to model? For instance, could they cycle around a thousand times, mutually modifying their states, and then eventually arrive at the state where they change the external or internal states somehow?

So that means my question is, if they mutually are able to change inside these Markov blankets the states without any trace in the external or internal states, then they can catapult or manipulate themselves to come into a situation where suddenly they produce a trace.

So they produce a trace. And if this is allowed, then I'm wondering what are we supposed to model with such processes.

So because, I mean, you can explain any arbitrary behavior and is it Turing complete, this kind of machine that we are then trying to implement there?

And basically, doesn't it undermine the explainability of behavior?

um so I guess you got my uh my question but maybe I'm understanding something wrong there anyone else want to give yes please Ali and then go on

I believe those uh bi-directional paths between uh active states and sensory states are uh simply there for the model to be uh more comprehensive and uh they basically refers to uh they basically refer to uh something like reflex arcs uh which don't necessarily need uh to pass through cortical paths uh so that's something uh much more directly manipulated uh but of course those um direct paths between active states and sensory states could be removed from the diagram and uh we we can talk about bypassing internal states or external states when the reflex arcs happen but I believe those direct paths are there to account for those kinds of uh intrinsic or some somehow uh those um uh the behaviors akin to reflex arcs thanks Ali but but isn't this isn't this then uh the question of what you define to be internal states if you say the internal states only start

up from a certain point above the spine and below it's no more internal then it's an arbitrary decision right so you could just also say that everything that happens in the hypothalamus is uh not part of internal states and only what is in the prefrontal cortex is and I mean how um I mean this is arbitrary you know and if what is the reason to have such kind of arbitrary decisions there you could then put everything into into this direct uh interaction or depending on where you yeah yes actually uh it is uh to some degree uh an arbitrary uh boundary because uh as we also read in chapter six uh defining the markov blanket boundary depends on the situation we're trying to model uh so uh it's not necessarily something predefined and it depends what behavior we're trying to model so uh yeah in some sense uh it depends on the context uh and yeah but but see my point is if we are trying to model reflexes we would put this into the internal states but if we are not trying to uh i mean i mean if you allow this direct interaction between active states and sensory states you're basically isn't this just like saying there is behavior that we cannot explain but we want to have it in our model oh i will give a thought but first jacob

yeah i was um i just wanted to add that you could i i think i agree with ali's point that it's probably there for um more i guess explainability or just to kind of um point to the fact that the active states and sensory states form like a single blank uh blanket but i think you could also think of it as enabling another loop with the internal states perhaps where if the external states are very high dimensional and you're in a partially observable state where you have like a window of observation then the active states can directly change the sensory states without necessarily changing the external states it's like um saccadic eye movement it's not changing the external states but it is changing your window of observation through action and i think that um carl mentioned in one of his talks that um these arrows might also represent different levels of sophistication but i'm not totally clear on um on how that uh can be explained

thanks okay few um angles on this so i think the general uh modeler perspective is what some call an arbitrary decision others might call a modeler degree of freedom so the idea that just because something is subjective or arbitrary devalues it rather than that is the space in which decisions are made and so it's rarely a bad thing to have a broader family to choose from and then um let's imagine that there is

okay second point is the particular partition which is this one so here's the particle

the blanket and the internal states and then the autonomous states which are the ones that we're most interested in developing imperatives for are internal states and active states so those are called the autonomous states they're the ones like if you had total control of your mind and action and you had the optimal policy like you'd be thinking and acting optimally and then that leaves the sensory states as being what we're trying to reduce the divergence with in terms of what we actually observe and what we expect slash prefer okay so the particular partitioning comes into being or a defined set of variables in a Bayes graph it's not like one variable is simply internal states so depending on what model you construct

there might be a variable might be internal with respect to one but then blanket with respect to another
 so if we were doing um an i circading model like you you might be able to make a Bayes graph
 where different things corresponded to differences um and then also this allows us to have a model
 where the entire cognitive behavior is accomplished by the blanket so you could have a model where these
 edges with internal states do nothing and that is corresponding to the the strange types
 that we talked about previously like it corresponds to an inert particle that's not doing anything that we
 might call truly cognitive which is to say that its perception and action are mediated entirely by its blanket
 and so there are systems where the entire performance can be understood on the holograph
 on the blanket other systems we might want to extend that and there might be a system where this edge
 is parameterized from empirical data to be zero or where you set this to be zero
 but another way to view this
 um graph
 is as a four by four matrix
 where the edges reflect relationships
 so if it was just four independent variables
 it would look like the identity matrix
 matrix and then the off diagonal elements of that four by four matrix reflect the causal influence of
 different variables on each other
 and so you can have the around the clock you could have the bi-directional around the clock
 you could have the telepathic model where there's a direct edge between internal and external
 so you could fit the full four by four but that would be equivalent to fitting a linear regression with four
 variables testing for all by all interactions which you might want to do
 also that has the least statistical power and so in terms of which edges it makes sense to
 to fix to zero removing the telepathy slash telekinesis edge is a really important one
 and then there's a few other edges that are removed like internal backward causing sense and external
 backwards causing action but that's not to say this is the only topology of the action perception loop
 michael
 yeah so i'm just wondering how well i as i as i said no i'm trying i'm i'm trying to
 apply this to our translation process data so we have a gaze data and keystrokes and so on
 and there is one observation in a translation that it's a mixture of of priming processes so of
 partially automatized processes and monitoring processes where higher cognitive decisions or
 reflective thought maybe comes into play so means that the translation is a mixture of automatized
 translate production routines and monitoring intervention and i'm wondering whether this model then would
 be a word um be could be used to say that these priming processes are the direct links between sensation
 and action
 and um and and and these higher order reflective processes and so somebody maybe thinks of in which
 context
 does the translation make sense and all this whether this would then be better modeled with the other route

so um so so in this case of course these priming processes happen in the head somewhere it's not in the spine right so translator produces and nevertheless uses the head but there's priming processes and other kinds of processes do you think that makes sense to view it that way

i think we can actually very quickly run the recipe and understand ali first and then we'll come to the recipe um i think one point that might help to understand this diagram slightly better is that uh in most uh most uh most of the literature uh related to active inference uh nfp uh the word particle is used uh to refer to the joint set of internal states and blanket states but in some uh other papers uh instead of particles they use system or agent so uh actually in some sense those markup blankets uh belong to the to the agent we're uh we're trying to model uh but uh the reason for behind distinguishing between the markup blanket and internal state is purely uh for modeling purposes uh i'm not sure if it makes sense uh or not but uh yeah so in any case those markup blankets would actually constitute parts of uh the sentient agent we're trying to uh model here

okay thank you let's just imagine just rapid fire you have the most translation um experience but others can like imagine um potentially what a translation setting would be so we'll just give one thought we don't need to um expound on all of them but we're gonna focus on the particular partition and then we'll use that to understand like what family of action perception loops are we actually fitting so what question are we looking to answer or what phenomena are we looking to model

are you asking me sure what's an interesting question for you or for anyone about translation human translation

yeah so we would like i guess try to understand translator environment interaction and how translators produce translations what are the difficulties what are the processes that goes on go on um difficulties okay which data do we have

we have the observations on the markov blanket so we have gaze data reading patterns on the source and the target text and we have typing behavior so the insertions and deletions that happen in the texts okay

all right so now we're gonna go to the form of the generative model as we mentioned last week like it doesn't have to necessarily be in this order each time but now we're going to so here no markov blankets need to be mentioned this is just empirically which data exists or what are we trying to model now let's think about um using the ontology what are the particular partitions going to be so

okay

the action states

the action states these ones

do correspond to actions

that you you have collected on

so we have icicades

which there's actually there are active models of um sentence comprehension
and typing

so here the icicades can be understood as being selected from directions
that the eye can move

and the typing could be understood as like let at the letter or the word scale

okay

what sense data

or sense states are going to be incoming

well i would i would i would put this icicades into the sense

thing or the reading pattern so we with an eye tracker we can see

uh where the eyes are on the text so we can see the um the items the tokens that are uh red so yes

so the sensei could be the incoming visual perception which token is um being viewed

but the the icicade is an action

yeah but okay

this is a policy up down left right that the eye can move to obtain a different sense state

okay

so

they're describing random variables in a bayes graph

they're not necessarily describing

um spatially separated parts of an organism

okay okay now this is where there's a lot of degrees of freedom in what is being modeled as the internal the
external states

so does anyone have a thought on what the internal or the external states might be

so the way we we are trying to see this is that the internal states

can be fluent so the the person can be in a steady state so they can be

just looking a little bit ahead fluently translating so it's a kind of a

fluent production state

uh exactly non-equilibrium steady state or they can be in a state where they are confused

or hesitating we can see this by the eye gaze data going back and forth in the text

revision trying out some options and so on so there's kind of a hesitation

that's what be an internal state in my understanding searching and

uh well a kind of surprise

or they could be in a state where they try to so what i think it's called um epistemic

state whether where they go ahead intentionally and try to you know activate the

correct or the appropriate mental resources to be able to then um use this in in a in a

try in a translation phase in a steady state does that make sense okay i i might put forth that

fluency confusion hesitation or just thinking these are like categorizations that you're going to be able

to delineate from the phase space of a variable and so the question is what is the variable so one option but anyone feel free to give a different option would be the external state are the true hidden semantics of the text and the internal state would be the received semantics of the text like just thinking about um numbers like we're or you know we're we're doing a cooking recipe but we're less fluent or we're translating a cooking recipe so it's three cups of sugar and two cups of flour we're going to be engaging in icacades to reduce our uncertainty about the semantics of a text but of course we can't directly see the semantics of a text what do we get incoming visual tokens what do visual tokens do visual tokens reduce our uncertainty about the semantics what do we do now that we've resolved semantics well we can take actions which constraining or if we didn't have to type this is a complex because there's like two kind of very different kinds of actions but let's just say we were just reading to understand so we're doing silent translation we're just reading and understanding something in a in a language so then if you had sufficiently resolved your uncertainty about us about the the word you'd imagine that you would take an icacade to where you expect the next piece of semantics to be resolved next word whereas if the incoming sensory states are not adequately resolving your uncertainty you might pause or you might even go back and then from these from patterns of flow across the the the agent states then you could categorize those into categories like fluent reading hesitant reading distracted that's one option or one can imbue the model with like a little bit more like a more opinionated model where you could say an internal state is going to be whether they're in this or that mode so just to kind of summarize that you can have internal states reflect semantics and then have a secondary categorization of different zones in that phase space or you can have a more opinionated model that is explicitly modeling transitions between for example fluent production and confusion i need to think about this have a good ideas thank you yes yeah and then just just to kind of let's just continue just briefly to these last piece um how would we set up the generative model what are the variables and priors okay so here was just natural language so the semantics of the text so what should that variable be are we reading a text that's just ones and zeros are we doing a text that's just one of the characters or is it is it is there going to be are we doing genre detection and there's five genre or what is the actual state space of different variables again for for typing are we going to have the state space be letters and then basically have like a transition matrix of letters which is like some of the earliest information theory or will we have a transition matrix of words or a transition matrix of motifs so those are all what some might call subjective but they're model or degrees of freedom there isn't a right answer there's just what model is made we we have actually uh something we call activity units so we have we can chop up the text into all the translation process into smaller pieces where we can see whether the person types or reads a text or has concurrent typing and reading and so we can chop up the timeline into smaller units which we call activity units and um and then and this i think would be the units on which we would like to um work

where so the idea here is not to simulate a translation system we're we're not so much interested what is actually the text the translation being uh produced but or say the textual elements but rather the process elements so means how long does it take what is the interval between successive keystrokes and what is the what is the offset between gaze and typing typing and so on what are the gazing patterns so how how long ahead do people read in the text uh and so on so so it's more the temporal structure that we want to model not the um not the semantic structure actually yes great so we are not trying to simulate a machine translation system yep so this so this could be simply like the most coarse-grained typing model would be there's two affordances typing or not typing yes yeah and and again one person could say these are subjective differences between models but these are the kinds of models if one were to really do a research project on that you could compare these three models or you could say we chose to pursue a model with this state space for this reason or like which parts are fixed and which must be learned well we know that there's things that can be learned by real humans but the question is what is going to be fixed or learned in this model

and fixed models are simpler computationally and they have a simpler interpretation but sometimes the function of learning then gets displaced and results in

in distorted behavior from other parameters

so this is why modeling is an iterative process and why it involves constructing sometimes vast families of cousin models to understand the stability of different parameters

because maybe once you enable learning on something

then that might change another parameter because all of a sudden it doesn't need to be taking on some kind of role

and then

which aspects are perception like

which aspects are learning like

and then the last question is like how to set up the generative process

interesting little hidden character

what are the elements

not sure if that's just my reader but that's very interesting

what are the elements of the generative process and how do they differ from a generative model

so the generative process

the niche it could be another active agent

or it could just be

a static text

that passes tokens in order

or it could be a conversant

but the exact structure

of the state space

and then which families of statistical distributions one chooses for priors

according to their expected free energy

the continuous time case is structurally similar which is the big visual message of figure 4.3

but note that while b in the discrete time case is a transition matrix how does the time point move into the next time point so you multiply forward with b and then that's how the hidden states move forward and that's where policy intervenes how hidden states change through time

in the continuous time case it's a lot more like a Taylor series expansion

because node 3 here is the derivative of x conditioned upon x and v causes

so in this case it's not that we're explicitly predicting at five time steps from the future if i do this or that

but rather under a given policy that's being considered

there's a function that's being approximated with higher and higher derivatives

so it doesn't have as straightforward of an interpretation you still could ask under this Taylor series approximation

at five time steps out what's my prediction so you can ask that and you could even compare under u^m derivative

uh under v_1 or v_2 two two different ways of taking derivatives we can evaluate so it's not that you can't evaluate

the u^m alternate futures contingent on policy selection

and that is exactly what happens with expected free energy in continuous state space models

so it is possible to compare alternate futures and still use expected free energy

but only in the discrete case do we actually have sequences of actions defined along the way

so does this mean if in the non-discrete case

we need to look into the seventh derivation to see what happens say seven points in time ahead

no in there's a great question u^m you could do u^m you could do a Taylor series uh a depth of one

so a Taylor series depth of uh with approximation of one let's see if there's a graph okay

okay

so let's just say we do a Taylor series with one term

so it just is literally a flat line

we can look ahead any depth

we could look ahead a thousand time points with a Taylor series a depth of one

and we don't need to explicitly predict those thousand time steps

whereas if we wanted a thousand time steps deep with discrete we would need to explicitly model every single step what is the b matrix from 997 to 998

so that's the strength and the weakness of the Taylor series

the Taylor series you can have a really simple u^m approximation

oh well how about two terms let's do the green one

and you could use that two term approximation

any depth

and as you get u^m more and more terms

they just like principal component dimensions
they always they always make a better and better approximation
sometimes a lot sometimes a little but you always do get better
so kind of like the contact zone
between the true function blue cosine in this situation
the contact zone
as you do higher approximations
always extends out more
like
brown peels off here
purple peels off here
and that's part of the issue is also with taylor series
and just modeling in general
you don't really know how well you're doing
because the um
the green line is not going to complain
oh what's the what's the cosine approximation of 11
that's negative 200
no
but the model's not going to complain
Ali?
possible unmute Ali
thank you
uh first of all sorry i have some connection problems
uh i'm not sure if my voice is
uh intelligent but uh one interesting uh point is
uh actually some uh neuroscientists such as
uh Rizzolatti and Senigalia
two prominent italian neuroscientists
uh explain this
choice of actions
in terms of mosaic of zones
so each zone
in this mosaic
would contain information
about uh the parameters
that must be used to select the most
appropriate variant of action

uh to act on an object
or
okay
okay
we we did just lose
um Ali but
we got that part
mosaic of zones
um
okay
let's see if we can get
but um
so suffice to say
discrete state space
downside
you have to explicitly model
you know
the upside
you get to explicitly model
downside
you have to explicitly model
so if you want to do temporal depth of 30
you need to run that out
across the combinatorics of policy
for a depth of 30
so the computational complexity
can be extremely high
which we explored in the branching time
active inference discussions
so here is like it makes a massive
tree search
so it's kind of like a chess algorithm
at least some chess algorithms
continuous state space
has a very very defined
uh computational complexity
however
it doesn't have the exact

um
ability
to say like
other than just generating this smooth function
you can't really say too much more
like how would it have been different
if I went up up up up up down
versus
up again
okay
we um
Ollie we heard you with a mosaic of zones
so you could maybe
um
add the links
into the page
any one or two more questions
that people would like to
come to
how about this one
what is meant by a variable
states or observations
how can states be continuous
so states can be continuous
a continuous variable
is like one where there's a smooth knob
that is controlling the value
so this is like a smooth knob
could be any number between 0 and 1
could be .77772
or discrete could be like 0 or 1
or it could be 1, 2, 3, 4, 5
so that's how states can be continuous
what is meant by variable
states or observations
so variables and states
basically mean the same thing
like a variable is just a

it's a parameter
in an equation or in a program
and variables take on states
so the variable could be continuous
between 0 and 1
and then the state that it's in is .6
and then observations
are one kind of variable

Michael?

Yeah, so how would you then
model the difference between
the observations and the states?

Is there a problem?

Is that straightforward?

So if I have a knob
and I can turn it on and off
so I have two observations for that knob
or I can turn it
you know like a dimmer or something
with infinite number of observations
but the knob itself
is just one thing
so I'm not sure

Yeah, great

I should understand that

Okay, so we're doing a volume knob

The knob is the territory

It has like physical

you know, it's made of metal

Okay, now there's multiple maps
that you could make of that knob

You could ask whether it's playing or not

That would be a binary discrete space

You could say

we're going to do

10 volume levels

This would be discrete

with 10 possible states

or you could do 100

or you could do a continuous variable

Any number from 0 to 1

So it's like a

it's a total classic

map territory question

State spaces

are about variables

Variables are in the map

If I make a variable

corresponding to weight

like of a person

it could be

over under 100 pounds

It could be

integers

It could be

tenths of a pound

That's still a discrete state space

It could be a continuous

number

Jonathan wrote

You could then imagine

that observations could be discrete

even when states may be continuous

It may be that we can't tell the difference

between the very finely graded states

Yes

Great point

Like just noticeable difference

And

when you do

discretized models

it becomes like

a trade-off

where

the more coarse graining

you do

the simpler the model is
but you're making bigger
and bigger buckets
that are clumping together
more and more
disparate things
Whereas if you do this
really fine scale
discretized model
like
hundredth of a pound
and we're going to be
weighing people
then you can get
all these
like bizarre
phenomena
where it's like
well if they're
137.1 pound
then this is the case
but if they're
137.2
then this is the case
So finding
that optimal
discretization
is a big
topic
in optimization
in general
and in
modeling
and then another
piece to remember
the state space
of the external state
so we're still not even

talking about the territory
itself
but the state space
of the external
variables
so our
model of temperature
it doesn't have
to be the same
as the cognitive
model of the agent
so we could say
in the simulation
temperature
is an integer
and then
they're going
to have
three states
cold
just right
and hot
or
here
in the external
world
it could be
continuous
variable
and then
in the brain
it could be
three state
or it could be
a continuous variable
or it could be
anything else
what matters

is
the observation
coming in
and then how
that gets fit
into the generative
model
Ali
yeah apologies
I got disconnected
as I was saying
some neuroscientists
explain
the choice
of appropriate
action
in terms of
the mosaic
of zones
and by that
they mean
that each zone
is the mosaic
that would contain
the information
to act
upon
the affordances
or the goal
directed
sequences
to act upon
the affordances
so
as an example
take for instance
the task
of picking up

a cup of coffee
this seemingly

simple task
can be done
in various ways
depending
on whether
we are drinking
from it
in that case
which we would
probably
grasp it
by the handle
or
if we're
washing it
so in that case
we would probably
grasp it
by the rim
or moving it
out of the way
which would
which in that case
grasping it
by the body
would probably
be the more
appropriate
sequence of action
so which of these
actions is deemed
appropriate for a task
is worked out
within that
mosaic of zones

which
they also
correlate
with some
cortical areas
which
maps
those
mosaic
zones
quite literally
in order to
afford
the agent
this
ability
to pick
the appropriate
action
in each
situation
thank you
cool
all right
that is our
second
discussion
on
chapter
six
so
six
in summary
outlines
some of the
most important
design choices
that have been

made in setting
up an active
inference model
we have
explored a lot
of auxiliary
routes
and questions
and hopefully
this conveys
some of the
richness
and excitement
and openness
around making
a model
because even
for simple
phenomena
there's many
ways to do
it
and even
for simple
phenomena
it's not just
about making
one model
and parameter
it's like an
iterated
modeling process
where you're
generating
these families
of models
they provided
a recipe

some guidelines
and thoughts
they point out
that it doesn't
have to be
followed
in that
particular
order
this sets
up the
remainder
of the
book
which puts
the ideas
into practice
through a
series of
illustrative
examples
designed to
showcase
the theoretical
principles
presented in
the first
half of the
book
in everything
that follows
the only
differences
amongst the
example
rest on the
design choices
we have

highlighted
part two
illustrates
systems with
different
boundaries
discrete or
continuous
dynamics at
different
time scales
and that
will all be
implementing
act
inf
so
we're going
to head
into
chapter
seven
next time
chapter
seven
is active
inference
in discrete
time
we're going
to look at
generative
models that
have a
discrete
time
characteristic
chapter

eight
is
continuous
time
so that
distinction
was kind
of raised
earlier
in figure
4.3
chapter
seven
and eight
are like
a pair
that are
going to
highlight
those two
different
possibilities
chapter nine
is about
data driven
analysis
and chapter
ten is a
conclusion
chapter
so
that
concludes
this
session
thanks
everybody
for the

great times

and we'll

come back

next week

for chapter

seven

thank you

thank you